

Unit III

PHYSICAL FITNESS

Introduction

Physical fitness is just one aspect of overall fitness, which also includes mental, emotional, and social well-being. Achieving physical fitness depends on several key factors:

1. Access to reliable medical and dental care.
2. Maintaining a balanced and nutritious diet.
3. Ensuring sufficient periods of rest and relaxation.
4. Engaging in consistent physical activity or exercise routines.

However, relying on any single element alone cannot meet the comprehensive demands of our society. Physical fitness is dynamic rather than static; it diminishes when individuals stop their regular exercise routines and physical activities. The effectiveness of physical activity is also dependent on the interaction with other factors that contribute to overall fitness and well-being.

General Objectives

At the end of the unit, the students are expected to:

- Understand the concept of physical fitness as the ability to perform daily tasks efficiently without undue fatigue and the capacity to enjoy leisure activities and meet emergency demands.
- Differentiate between health-related components (e.g., cardiovascular endurance, muscular strength) and performance-related components (e.g., agility, coordination) of physical fitness.

- Explain concepts related to physical fitness: Students will grasp key concepts such as endurance, strength, flexibility, and agility, and their significance in achieving overall physical fitness.
- Discuss benefits of physical fitness: Students will analyze the numerous benefits of maintaining physical fitness, including improved posture, reduced risk of injury, enhanced mental resilience, and increased vitality.
- Apply fundamental principles such as specificity, progressive overload, and recovery to design effective fitness programs tailored to individual needs and goals.

The Meaning, Concepts and Components of Physical Fitness

Physical Fitness is the ability to perform one's daily task efficiently without undue fatigue but with extra "reserve" in case of emergency. Being *physically fit* means being able to perform one's daily task efficiently without undue fatigue and still have extra energy to enjoy leisure activities and / or meet emergency demands.

A state of overall well-being achieved through regular physical activity, proper nutrition, and adequate rest. When someone is physically fit, they not only manage daily responsibilities efficiently but also cultivate mental resilience and emotional stability. This holistic approach to fitness promotes longevity, enhances quality of life, and fosters a proactive approach to health. It empowers individuals to not only cope with stressors but to thrive in various aspects of life, whether pursuing hobbies, engaging in recreational activities, or meeting unexpected challenges head-on.

Thus, physical fitness is not merely a goal but a lifestyle that supports optimal functioning of life.

Based on this definition, there are (3) three important aspects that an individual should be able to meet to be considered physically fit. These include:

1. Being able to perform one's daily tasks without getting too tired before the end of the day. These daily activities include:

- a. Waking up in the morning to prepare for school or work
 - b. Going to school or office on time
 - c. Attending classes regularly
 - d. Doing assignments in the library
 - e. Meeting other individual, classmates and peer groups
 - f. Going home
 - g. Doing homework and household chores
2. Enjoying leisure time in some form of recreational activities. Leisure time is the amount of time that is left after the daily routine activities of an individual have been accomplished. Recreation, on the other hand, refers to any activity that is participated in by the individual during leisure time on voluntary basis because it provides immediate satisfaction to the individual. Recreation takes place in the form of:
- a. Watching TV or movies
 - b. Talking to friends over the phone
 - c. Reading books and magazines articles
 - d. Doing arts and crafts
 - e. Listening to radio
 - f. Playing sports
3. Meeting emergency demands such as:
- a. Doing important errands
 - b. Performing social obligations
 - c. Attending to visitors
 - d. Attending to household problems

Concepts of Physical Fitness

1. **Organic Vigor**- refers to the soundness of the heart and the lungs, which contributes to the ability to resist diseases.

2. **Endurance-** is the ability to sustain long-continued contractions where a number of muscle groups are used; the capacity to bear or last long in a certain task without undue fatigue.
3. **Strength-** is the capacity to sustain the application of force without yielding or breaking; the ability of the muscle to exert effort against the resistance.
4. **Power-** refers to the ability of the muscle to release maximum force in the shortest period of time.
5. **Flexibility-** it is a quality of plasticity, which gives the ability to do a wide range of movements.
6. **Agility-** is the ability of an individual to change direction or position in space with the quickness and lightness of movement.
7. **Balance-** is the ability to control organic equipment neuro-muscularly, a state of equilibrium.
8. **Speed-** is the ability to make successive movements of the same kind in the shortest period.

Components of Physical Fitness

The components of physical fitness are divided into two categories: the *health-related components* (flexibility, cardiovascular endurance, muscular strength, muscular endurance and body composition) and the *performance related components* (agility, balance, coordination, power, and speed).

I. Health-Related Fitness

Health-Related Fitness- is a function of body's adaptation to exercise. It could be developed and maintained through a regular and proper exercise program. There are at least five (5) components of health-related fitness. They are as follows:

- A. **Muscular Strength** refers to the ability of the muscle to exert maximum effort in brief duration. It may be developed through isotonic, isometric, or isokinetic contractions.

1. **Isotonic Contractions** are voluntary contractions in which muscles shorten and lengthen alternately. The muscle contracts at varying speed against a resistance. These contractions are of two types:

- a. **Concentric Contraction**- refers to muscles that shorten during exercise. There is no movement that occurs, and the length of the muscle remains unchanged.
- b. **Eccentric Contractions**- refers to the muscle that lengthens during an exercise. Muscles exert force due to outside resistance.

The most common example of isotonic contraction is *calisthenics exercise*. This makes use of a body segment or the entire body weight as a form of resistance during exercise. Some of the most traditional calisthenics exercises that are being performed to increase muscular strength are push-ups, sit-ups, and pull-ups. Another approach to develop muscular strength involving isotonic exercise is using weights.

2. **Isometric Contractions** are another type of contractions in which the muscles are contracted against an immovable resistance. The exercise provides maximum contraction of muscle; however, there is no observed movement during the exercise. This is done in about 5 seconds.

3. **Isokenetic Contractions** are like isotonic contractions, but the muscles are exposed to fixed machines with variable degrees of resistance. Through the range of movement, resistance is equal to the force applied.

Two physiological benefits derived from strength training:

1. Increase muscle strength by 10-25% within 6-8 weeks; and
2. Increase muscle size or muscle hypertrophy.

B. **Muscular Endurance** refers to the ability of the muscle to endure a sub maximal effort for a prolonged period. Performing a strength exercise such as push-up over an extended period of time or performing the same exercise for many continuous repetitions is a demonstration of muscular endurance. One

basic requirement in achieving muscular endurance is muscular strength. Muscle endurance is developed by employing the different strength training exercise mentioned above either for longer periods of time or an increased number of exercise repetitions.

- C. **Cardiovascular Endurance** refers to the ability of the heart, blood vessels and the lungs to adapt to physical exertion for a prolonged duration. Usually, the heart rate is increased from the resting pulse rate to the target heart rate that the individual has to sustain for a period of 20 to 60 minutes. With high levels of endurance, an individual has more energy and can sustain distinct types of activities. It is also likely that heart disease (CHD) is prevented.

There are important variables to consider when engaging in any cardiovascular endurance program. These are:

1. Intensity which refers to how stressful exercise is.
2. Duration which refers to how long will the exercise be performed.
3. Frequency refers to the number of times the individual will exercise each week.
4. Mode which refers to the kind of activity selected such as brisk walking or jogging.

Physiological Benefits of Cardiovascular Training Program

1. Decreased resting heart rate
2. Decreased recovery time from a bout of exercise
3. Increased blood volume and red blood cells to transport oxygen throughout the body
4. Stronger heart muscle allows it to eject more blood (stroke volume) with each heartbeat. If this occurs, the heart can rest longer between beats
5. Increased number of capillaries for efficient exchange of oxygen, carbon dioxide, and nutrients between blood and body cells

6. Decreased rate of respiration
7. Increased aerobic capacity which allows the body cells to utilize oxygen for better work efficiency.
8. Prevent and/or reduction of the risk of developing coronary heart diseases.

Activities that develop cardiovascular endurance

- | | |
|----------------------------|------------------------|
| 1. Prolonged brisk walking | 6. Continuous swimming |
| 2. Prolonged jogging | 7. Rowing |
| 3. Stationary bicycling | 8. Aerobic dancing |
| 4. Prolonged skipping rope | 9. Hiking |
| 5. Playing basketball | 10. Playing football |

D. Flexibility is the ability of the muscles and joints to go through a full range of motion. Flexibility reduces the risk of injury, enhances performance, and prevents muscle soreness. It is achieved through stretching designed to lengthen or elongate shorten soft tissue structures and thereby increase the range of motion. Flexibility is influenced by three factors:

1. Structure of the joints
2. Number of tissues surrounding the joint
3. Extensibility of the ligaments, tendons and muscle tissue that connects the joints.

It involves four basic movements, namely: **flexion** (bending of a body segment), **extension** (straightening a body segment), **abduction** (moving a limb away from the body), and **adduction** (moving a limb toward the body).

Fitness Benefits of Flexibility Exercise

1. Increased range of muscle joint motion;
2. Reduced muscle stiffness and increased body relaxation;
3. Improved blood circulation in specific body segment;
4. Reduced incidence of injury during a major sport event; and

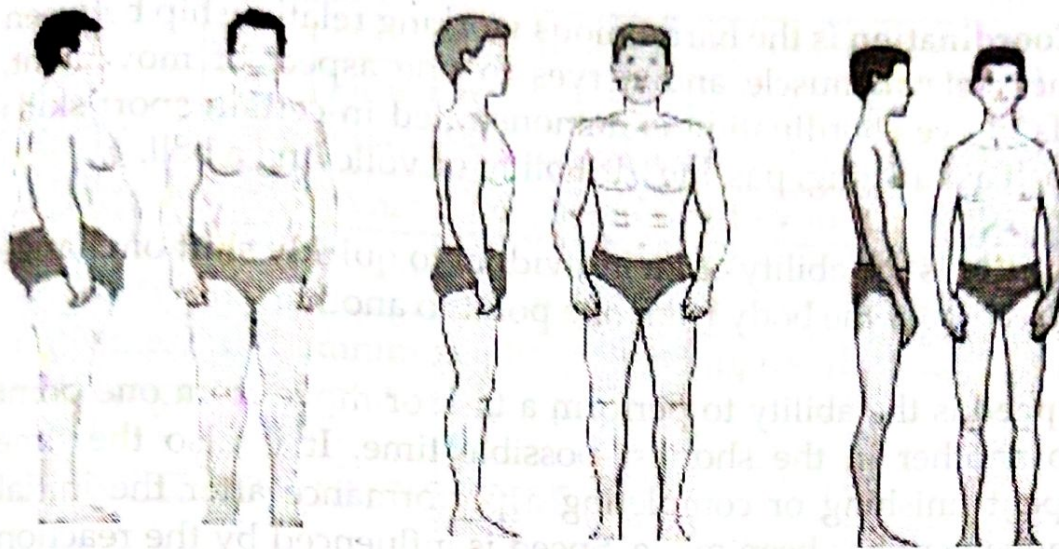
5. Reduced risk of cardiovascular problems in exercise.

- E. **Body Composition** refers to the proportion of lean body mass to fat body mass. It stresses one's relative fatness or leanness in relation to height. In recent years, body composition has gained wide acceptance in exercise science as one of the components of health-related fitness. Its inclusion was earlier opposed since supporters believe that fitness is for everyone, including the slim and overweight. In addition, being fat has some genetic predisposition that may be quite difficult to alter.

Somatotypes

Somatotyping or **body typing** is a system of classifying an individual according to the shape of the body. It was developed by Sheldon during the 1940's and 1950's. He noted that the physique of the body may be categorized into three distinct types:

1. An **ectomorphic** body type is characterized as lean and small body build with greater surface area to mass ratio. Bone size is relatively small with slender limbs and low muscle mass.
2. A **mesomorphic** body type has a relative predominance of muscles. The bones are large and heavy with massive limbs, thus contributing to greater weight than the ectomorphic body type.
3. An **endomorphic** body type is characterized by a relative predominance of soft roundness and large digestive viscera. There is a greater percentage of body fat when compared to lean body mass.



Endomorph

Mesomorph

Ectomorph

Somatotypes are of special interest to fitness enthusiasts and athletes. It helps the individual understand the extent of weight reduction or weight gain he/she expects to achieve given a well-defined exercise program. Thus, one of the factors that affect weight gain or weight loss aside from exercise and a regulated diet is the somatotype. Among athletes, somatotype is highly correlated to excellent sport performance depending on the type of sport event. For instance, it has been found that among track and field athletes, physical characteristics of those successful in shot put differ from those successful in marathon. This suggests that for one sport event, a specific body type may dictate one's superior performance over another.

II. Performance-Related Fitness

It refers to the quality of one's movement skill. It includes five (5) general components namely:

1. **Balance** is the ability to maintain equilibrium in relation to changes in body position. Balance can be categorized into static and dynamic balance. Static Balance is the ability to maintain equilibrium in a fixed position such as standing on one foot or on a balance beam. Dynamic Balance is the ability to maintain one's equilibrium while the body is in motion. Walking on a balance beam is one example of this type of balance.

strength, improved cardiovascular efficiency, and enhanced flexibility.

5. **Principle of Recovery:** Adequate rest and recovery periods are essential for optimal performance and injury prevention. Muscles need time to repair and grow stronger between workouts, and proper recovery strategies (like hydration, nutrition, and sleep) are crucial for overall well-being.
6. **Principle of Reversibility:** Fitness gains diminish when exercise is discontinued. The principle underscores the importance of maintaining regular physical activity to sustain improvements in strength, endurance, and flexibility over time.
7. **Principle of Individual Variation:** Recognizing that individuals respond differently to exercise stimuli due to factors such as age, gender, fitness level, and health status. Adjusting exercise programs to accommodate these differences ensures safe and effective training outcomes for all individuals.

Benefits of Physical Fitness

1. **Vitality.** Muscles are basic for all body action. They increase in strength with activity and deteriorate from lack of it. Fit muscles use less energy to perform the same task, leading to an increase in vitality.
2. **Posture.** A physically fit person is able to maintain his general postural alignment better than one with weak musculature. Good fitness as a program stresses the development of anti-gravity muscles, which maintains good abdominal wall and the arm and shoulder girdle.
3. **Relieves Lowback Pain.** A lack of physical activity has been found to be a major cause for some cases of pain in the lower back. Most adults who suffer from low-back pain are relieved from this pain by exercises that strengthen the back and abdominal muscles.
4. **Retards Aging Process.** Continued participation in regular exercises of the proper amount and severity is of considerable

value in postponing deterioration that usually takes place as a person grows old.

5. **Physical Fitness and Ability to Meet Emergencies.** The body that is accustomed to sedentary living habits can usually operate ineffectively even though it is operating at near maximum effort.
6. **Neuromuscular Skill.** The smooth, efficient coordination of the muscular system is improved as a result of regular participation in physical activity.
7. **Relaxation.** Overactive minds in underactive bodies often need physical outlets for accumulated emotional and muscular tensions that seem to be relieved by actions of the skeletal muscle.
8. **Improvement of Personality and Social Skills.** Participation in games and sports aids in improving the personality and in developing desirable social skills.
9. **Mental Fitness.** Because of the mental and physical relaxation that often results from physical activity, regular exercise is thought by many to be of considerable value in aiding the natural mental processes to function with increased efficiency.
10. **General Growth.** The physically fit person usually possesses a high degree of general resistance, which enables him successfully avoiding minor illness.

The Parameters of Physical Fitness

Many proponents of physical education have various ideas on how to measure physical fitness. Educators on physical education today separate the various disciplines into four measurable parameters, which are:

1. Muscular Endurance
2. Muscular Strength
3. Cardio-respiratory Endurance
4. Joint Flexibility

The above parameters are considered the hard core of physical fitness. Without strength, participation in physical activity is not possible and the fitness of the individual is limited. When muscles are weak or lack endurance, all movements are curtailed, since all movements depend on the strength of the muscle or groups of muscles.

A fit individual can maintain a lower heart rate during activity which can supply a greater volume of blood with each contraction of the heart and can recover more quickly after exercise than a less fit individual.

As the late president John Fitzgerald Kennedy said, *"Physical Fitness is not only one of the most important keys to a healthy body; it is the basis of dynamic, creative, and intellectual activity in that intelligence and skill can function at the peak of their capacity when the body is healthy and strong"*.

WAYS TO PREVENT INJURY DURING PHYSICAL EXERCISES

Preventing physical injuries during exercise is crucial for maintaining long-term fitness and well-being. Here are some effective strategies to minimize the risk of injury:

1. **Warm-up properly:** Begin each exercise session with a thorough warm-up. This increases blood flow to muscles, improves flexibility, and prepares the body for more intense activity.
2. **Gradual progression:** Avoid sudden increases in intensity, duration, or frequency of exercise. Gradually progress over time to allow the body to adapt and reduce the risk of strains or overuse injuries.
3. **Use proper technique:** Learn and practice correct exercise techniques under the guidance of qualified instructors or trainers. Proper form helps to target muscles effectively and reduces the likelihood of injuries caused by poor biomechanics.
4. **Wear appropriate gear:** Use footwear and clothing suitable for the type of exercise you are doing. This includes supportive

shoes for running or cross-training and protective gear for contact sports or high-impact activities.

5. **Stay hydrated and nourished:** Maintain adequate hydration and proper nutrition before, during, and after exercise. Dehydration and nutrient deficiencies can impair muscle function and increase injury risk.
6. **Listen to your body:** Pay attention to signals of fatigue, pain, or discomfort during exercise. Stop and rest if you experience unusual pain or discomfort to prevent aggravating potential injuries.
7. **Include rest days:** Allow sufficient rest between workout sessions to give muscles time to recover and repair. Overtraining can lead to fatigue, weakened immune function, and increased susceptibility to injuries.
8. **Cross-train:** Incorporate a variety of exercises and activities into your routine to prevent overuse injuries and balance muscle development. Cross-training also enhances overall fitness and reduces repetitive stress on specific muscle groups.
9. **Cool down:** After completing your workout, cool down with gentle stretching or low-intensity exercises. This helps to gradually decrease heart rate and muscle tension, promoting recovery.
10. **Seek professional guidance:** Consult with a fitness trainer, physical therapist, or sports medicine specialist for personalized advice and exercise programs tailored to your fitness level and goals. They can provide valuable insights to prevent and manage injuries effectively.

Name:		Date:
Group:	Class Schedule:	Rating
Instructor:		

Score sheet 3.2

Quiz

- I. **Multiple Choices:** Read carefully the statement. In the space provided, write the letter of your choice that best completes the statement or answers the question. No Erasures.

PHYSICAL FITNESS AND FITNESS COMPONENTS

- ____ 1. Physical fitness is the result of the following factors. Which is not?
 - A. good medical and dental care
 - B. adequate rest and relaxation
 - C. proper nutrition
 - D. irregular exercise

- ____ 2. Which of the following activities meet emergency demands?
 - A. watching TV or movie
 - B. performing social obligations
 - C. going to school on time
 - D. playing sports

- ____ 3. One can achieve this component of physical fitness if one can perform one's task efficiently without undue fatigue.
 - A. endurance
 - B. speed
 - C. strength
 - D. power

- ____ 5. The ability to use your muscles against a resistance is ____.
- A. muscular strength C. flexibility
B. cardiovascular endurance D. speed
- ____ 6. Body composition is determined by the amount of fat and lean tissue in the body, therefore it refers to the proportion between ____.
- A. fat weight and total body weight
B. fat weight and height
C. height and total body weight
D. total body height and weight
- ____ 7. Body flexibility is achieved through stretching specific segments of the body. Good flexibility is best achieved when exercises be ____.
- A. done once a week C. executed lightly
B. performed daily D. forgotten
- ____ 8. Bending is an example of ____.
- A. endurance C. flexibility
B. strength D. agility
- ____ 9. The least effective exercise for promoting cardiovascular endurance is ____.
- A. bicycling C. walking
B. weight-lifting D. swimming
- ____ 10. All of the following are health-related components of physical fitness, EXCEPT
- A. cardio-vascular endurance
B. flexibility
C. muscular strength and endurance
D. agility

_____ 11. The power needed to jump very high integrates two of the following factors.

- I. muscular endurance
- II. speed of muscle contraction
- III. muscular strength
- IV. degree of muscular flexibility

A. I and III B. I and IV C. II and III D. III and IV

_____ 12. Of the following, the component of physical fitness most conducive to longevity is _____.

- A. muscular strength
- B. cardiovascular endurance
- C. muscular endurance
- D. body composition

_____ 13. Agility requires one to move one's body in different levels and directions. Which combination of components is achieved here?

- A. coordination and balance
- B. coordination and speed
- C. speed and endurance
- D. coordination and power

_____ 14. All of the following are skill-related components of physical fitness, EXCEPT

- A. Agility
- B. Balance
- C. Coordination
- D. Flexibility\

_____ 15. This is the ability of the individual to maintain equilibrium about changes in body position.

- A. Coordination
- B. Power
- C. Balance
- D. Speed

BENEFITS OF PHYSICAL FITNESS

_____ 16. Which of the following is not a benefit of physical fitness?

- A. general health
- B. Relaxation
- C. Aging
- D. Posture

- ____ 17. Which of the following benefits of physical fitness is achieved when one values continued participation in regular proper amounts and severity of exercise as he grows older?
- A. retards aging process
 - B. improves personality and social skills
 - C. relieves low back pain
 - D. helps mental fitness
- ____ 18. When one participates actively in games and sports, one achieves the benefit of physical fitness.
- A. improves personality and social skills
 - B. Relaxation
 - C. relieves low back pain
 - D. Vitality
- ____ 19. This test is conducted twice a year where the initial test is administered at the beginning and end of the year.
- A. PE Test
 - B. Fitness Program
 - C. Physical Fitness Test
 - D. PTest
- ____ 20. Which of the following statements CANNOT help an individual to remain healthy and fit?
- I. Perform everyday strenuous activities
 - II. Be an antagonist always
 - III. Involves himself in vices
 - IV. Achieve a high level of physical fitness
- A. I and III
 - B. II and IV
 - C. II and III
 - D. III and IV

Instructor's Signature over Printed Name